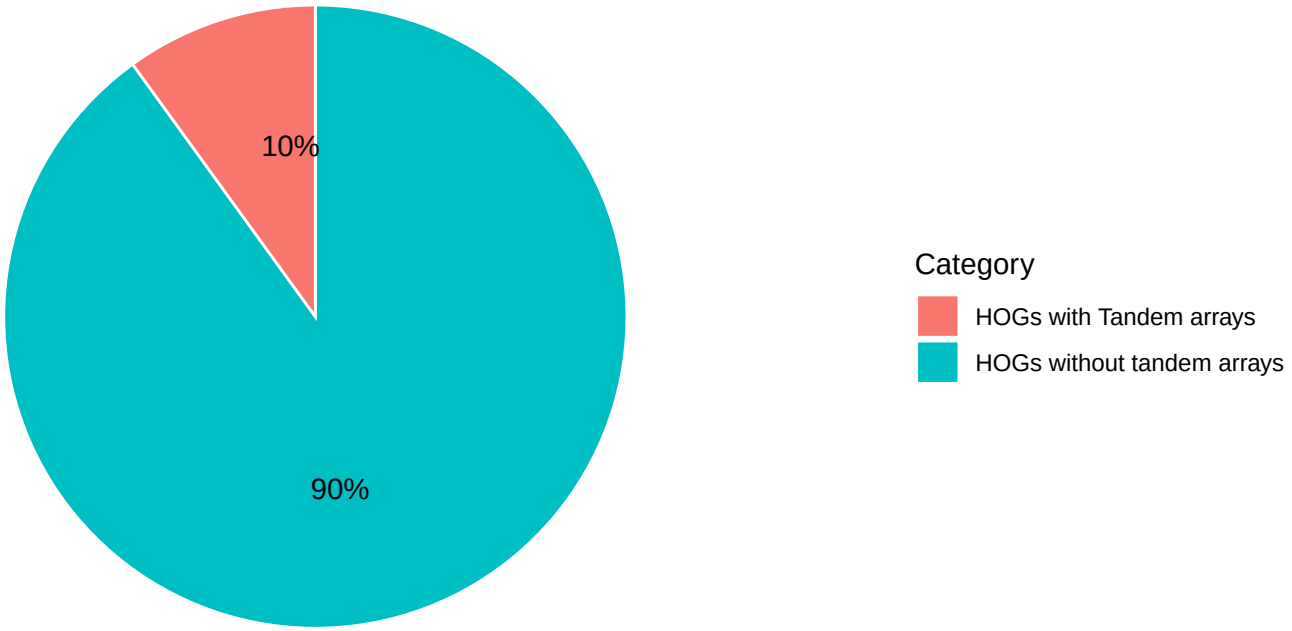Distribtuion of HOGs with tandem arrays